# Junyao Dong

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## Education

**Nanyang Technological University** — Singapore
M.Eng. by Research in Computing and Data Science Engineering — 08/2024 – 06/2026
- GPA: 4.17/5.0. Co-advised by Cheng Long and Shuhao Zhang.
- **Research Interests:** Vector databases, retrieval systems, and machine learning systems.

**The University of Manchester** — Manchester, UK
Visiting Student in Computer Science — 08/2019 – 05/2020
- Overall Average: 70% (First-Class Standard).

**Guangdong University of Foreign Studies** — Guangzhou, China
B.Eng. in Computer Science — 09/2017 – 06/2021
- GPA: 3.4/4.0.

## Publications (* = Equal contribution)

**The Program Counter Is a Path: Code, Data, and Control in One Addressable Tree**
Peng Li*, Junyao Dong*
- Submitted to the *ACM SIGPLAN Conference on Programming Language Design and Implementation* (**PLDI'27**). [Website] [Code]

**Shared Memory in Multi-Agent Systems: Which View Does the Agent Read?**
Junyao Dong, Xisen Wang, Yu Chen, Yi Li, Qi Zhang
- Submitted to the *International Conference on Autonomous Agents and Multiagent Systems* (**AAMAS'27**).

**ANNchor: Concurrency Control for Freshness in Streaming Graph-Based Approximate Nearest Neighbor Search**
Junyao Dong, Shuhao Zhang, Wenchao Zhou, Cheng Long
- Revision at *Proceedings of the ACM on Management of Data*, 2027 (**SIGMOD'27**, Round 3).

**CANDOR-Bench: Benchmarking In-Memory Continuous ANNS under Dynamic Open-World Streams**
Mingqi Wang*, Junyao Dong*, Zhuoyan Wu, Jun Liu, Ruicheng Zhang, Jianjun Zhao, Ruipeng Wan, Xinyan Lei, Shuhao Zhang, Bolong Zheng, Haikun Liu, Xiaofei Liao, Hai Jin
- *Proceedings of the ACM on Management of Data*, 2026 (**SIGMOD'26**). [Paper] [Code]

## Research Experience († = Full-time)

**Vector Database Group @ NTU** — Singapore
Research Associate† (Advised by Cheng Long) — 08/2026 – Present

Applying vector search and RaBitQ quantization to long-context LLM inference.
- Studying retrieval over the KV cache for sparse attention in long-context decoding.

**PageIndex** — Remote
Research Intern (Co-advised by Mingtian Zhang and Yu Tang) — 12/2025 – Present

Building retrieval over long, hierarchically structured documents.
- Led the design and implementation of ConDB, a chunk-free, vector-free context database for long documents.
- Collaborated with MongoDB on storing and searching large hierarchical JSON in the MongoDB Python driver.

**Vector Database Group @ NTU & IntelliStream @ HUST** — Singapore
Graduate Student Researcher† (Co-advised by Cheng Long and Shuhao Zhang) — 08/2024 – 06/2026

Investigated index freshness in streaming vector search, and how to benchmark it.
- Designed ANNchor, a concurrency-control framework that guarantees post-acknowledgment visibility on a single live HNSW graph; up to 4.6× median search throughput over global locking and 68% lower median end-to-end P99 latency under write-heavy load.
- Co-developed CANDOR-Bench, a continuous ANNS benchmark of 19 algorithms on 12 datasets under drift and concurrent read–write streams; co-designed the streaming evaluation harness.

**The Hong Kong University of Science and Technology (Guangzhou)** Guangzhou, China
Research Assistant† (Advised by Ying Cui) 04/2024 – 06/2024

Implemented adaptive video delivery for variable network conditions.

- Developed an adaptive bitrate system that cuts playback buffering without losing visual quality.

## Selected Open-Source Projects

**RaBitQ-Library** ∼300 GitHub stars
Contributor ◇ Code

Official library for RaBitQ, a randomized bitwise quantization method for approximate nearest neighbor search, covering quantization, scanning, and index integration (used by Meta, NVIDIA, Microsoft, and Alibaba).

- Hardened IVF index construction robustness and prototyped quantized graph-index search integration.

**RL-Kernel** 400+ GitHub stars
Contributor ◇ Code

High-performance RL post-training infrastructure for operator-level train–inference consistency.

- Implemented CUDA/Triton kernels for DeepSeek V4 compressed attention: one-softmax MQA with an attention sink, and grouped output projection.

**vllm-rlt** 200+ GitHub stars
Contributor ◇ Code

Inference engine for looped language models. Recurrent steps are batched continuously, so requests at different loop depths share one batch.

- Designed the engine loop so synchronous and asynchronous steps share scheduling and result handling, while submission, GPU completion, and output delivery stay separate.

## Industry Experience († = Full-time)

**Baidu** Shanghai, China
Software Engineer†, Cloud Services 05/2021 – 03/2024

Built live-video infrastructure, from edge ingest through storage to on-air production.

- Integrated H.264/H.265 at the edge for Baidu Cloud Live Streaming Service and wrote outputs as MP4, FLV, and HLS.
- Added watermarks, picture-in-picture, transitions, and subtitles to Baidu Cloud Live Cast Production Studio on a customized FFmpeg engine and Baidu's RTC stack.

**Momenta** Suzhou, China
Research and Development Intern, Autonomous Driving 10/2020 – 05/2021

Developed log infrastructure for replaying recorded drives as training data.

- Synchronized perception and planning streams onto one timeline.

## Awards and Honors

**Second Prize, National Modeling Competition** 2020

**Overseas Student Programme Award** 2020

## Academic Service

**Availability and Reproducibility Committee (ARC)**

- SIGMOD 2026 2026

**External Reviewer**

- VLDB 2026, SIGMOD 2027, VLDB 2027, WSDM 2027 2026
- ICDE 2026 2025